Hoogwan Lee

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Research Interest | *Advanced Packaging Integration, website: https://advanced-packaging-study-hg.vercel.app/*

Education

M.S. Department of Semiconductor Engineering *Expected Feb. 2027*

Seoul National University of Science & Technology, Seoul, Republic of Korea

- GPA: 4.5 / 4.5

B.S. Department of Electrical & Information Engineering *Mar. 2019 – Feb. 2025*

Seoul National University of Science & Technology, Seoul, Republic of Korea

- GPA: 4.24 / 4.5 (Major: 4.4 / 4.5)
- Academic Excellence Scholarship (5 Consecutive Semesters), 2022 – 2024

Internship, University of Nevada, *June 2023*

Las Vegas, US

- Exchange Student Program (one month)

Research

M.S. Research *Aug 2024 – Feb 2027*

Electronic Systems Laboratory, Seoul National University of Science & Technology

- *Advisor: Prof. Sarah Eunkyung Kim, Prof. Seulki Hong*
- Title of Thesis: Ar/CH_4 Reductive Plasma for Low-Temperature Hybrid Bonding
- Key Activities: DOE for Surface Optimization, a-C:H Kinetics Analysis, Post-Bonding Reliability

Capstone Design (Undergraduate Research) *Feb 2024 – Feb 2025*

Department of Electrical & Electronic Engineering, Seoul National University of Science & Technology

- *Advisor: Prof. Wonbo Shim*
- Title: Low Power High Integration 4T-SRAM Design
- Key Activities: MPW, Breadboard, TCAD

Publications

First-author

- **H. Lee**, J. Choi, and S. E. Kim, "Reductive CH4 Plasma-Induced Modification of Cu Surfaces for Low-Temperature Hybrid Bonding," IEEE Transactions on Components, Packaging and Manufacturing Technology, vol. 16, no. 4, pp. 880-889, April 2026, doi: 10.1109/TCPMT.2026.3665204.
- **H. Lee** and S. E. Kim, "Thickness-dependent Structural Evolution of a-C:H Layers Formed by One-step Ar/CH4 Plasma Reduction for Direct Cu Bonding," Sci Rep (2026). https://doi.org/10.1038/s41598-026-49285-6.
- **H. Lee**, J. Park, S. E. Kim, S. Hong, "Impact of Ar/CH4 Plasma Surface Modification on the Electrical Reliability and Mechanical Integrity of Cu Direct Bonding", IEEE Access, doi: 10.1109/ACCESS.2026.3727984.

Co-authored

- D. Lee, **H. Lee**, J. Choi, S. Jang, and S. E. Kim, "C_2H_4 Reductive Plasma Treatment on Copper Surface for Copper Bonding," Materials Research Express, vol. 12, no. 5, 055905, May 2025, doi: 10.1088/2053-1591/addc3c.
- S. Jang, **H. Lee**, and S. E. Kim, "Planarization of Cu Polymer Surfaces via Fly Cutting: Influence of Polymer Properties and Processing Parameters," IEEE Transactions on Components, Packaging and Manufacturing Technology. vol. 16, no. 4, pp. 890-898, April 2026, doi: 10.1109/TCPMT.2026.3667463.
- K. Seo, S. Kim, J. Choi, S. Jang, **H. Lee**, and S. E. Kim, "Critical Roles of Copper Surface Planarity in Achieving Reliable Cu/SiO2 Hybrid Bonding," IEEE Transactions on Semiconductor Manufacturing, doi: 10.1109/TSM.2026.3680409.
- B. Go, **H. Lee**, K. Komamura, T. Kishida, and S. E. Kim, "Electroless-Plated Gold Nanolayer for Low-Temperature Copper Hybrid Bonding," IEEE Trans. Components, Packaging and Manufacturing Technology, doi: 10.1109/TCPMT.2026.3687162.

Conferences

First-author

- (Poster) Simultaneous Surface Reduction and Self-Passivation via Ar–CH4 Plasma for Cu Hybrid Bonding, ECTC 2026, **H. Lee**, B. Go, S. E. Kim
- (Oral) Surface Reduction and Self-Passivation in a Single-Step Ar/CH4 Plasma Treatment for Hybrid Cu Bonding, ICEP-HBS 2026, **H. Lee**, B. Go, J. Park, S. E. Kim
- (Oral) Simultaneous Surface Reduction and Self-Passivation via Ar–CH4 Plasma for Low-Temperature Cu Bonding, ICOS 2026, **H. Lee**, B. Go, S. E. Kim
- (Poster) Advanced Cu Surface Modification Using Ar–CH4 Plasma Pretreatment for Direct Bonding Applications, ISMP 2025, **H. Lee**, B. Go, S. E. Kim
- (Poster) Effect of CH4 Plasma Pretreatment on Cu-to-Cu Bonding Applications, ICEPT 2025, **H. Lee**, D. Lee, S. Jang, J. Choi, S. E. Kim
- (Poster) Study of Low-Temperature Cu-to-Cu Bonding Using Reducing Plasma, ICEP-IAAC 2025, **H. Lee**, D. Lee, J. Choi, S. Jang, S. E. Kim

Co-authored

- (Poster) Electroless Noble Metal Passivation Enabling Oxidation-Free Cu Bonding Through a Comparative Study of Au, Pd, and Ru for Low-Temperature Hybrid Bonding Applications, ECTC 2026, B. Go, **H. Lee**, K. Komamura, T. Kishida, S. E. Kim
- (Poster) Analysis of Au and Ru Passivation Layers Deposited via Electroless Plating for Low-Temperature Cu–Cu Bonding, ICEP-HBS 2026, B. Go, **H. Lee**, K. Komamura, T. Kishida, S. E. Kim
- (Oral) Evaluation of Fly Cutting for Planarization in Cu/Polymer Hybrid Bonding and Its Effects on Polymer Materials and Processing Conditions, ICEPT 2025, S. Jang, J. Choi, D. Lee, **H. Lee**, S. E. Kim
- (Oral) Study on the Deposition and Low-Temperature Bonding of PVD SiCN Thin Film, ICEPT 2025, J. Choi, S. Jang, D. Lee, **H. Lee**, S. E. Kim
- (Poster) Study on Cu Dishing in CMP Process and Its Impact on Hybrid Bonding by Pad Size, ICEPT 2025, S. Kim, K. Seo, J. Choi, S. Jang, D. Lee, **H. Lee**, S. E. Kim
- (Oral) Effect of Palladium Passivation via Electroplating and Electroless Plating on Low-Temperature Cu-to-Cu Bonding, ICEPT 2025, D. Lee, J. Choi, S. Jang, **H. Lee**, K. Komamura, S. E. Kim
- (Poster) Effect of Noble Metal Passivation Deposited by ECD on Cu Surface for Low-Temperature Cu-to-Cu Bonding, ICEP-IAAC 2025, D. Lee, J. Choi, S. Jang, **H. Lee**, S. E. Kim
- (Poster) Exploring the Potential of Fly Cutting for Polymer Planarization in Cu/PDMS Hybrid Bonding, ICEP-IAAC 2025, S. Jang, J. Choi, D. Lee, **H. Lee**, S. E. Kim
- (Poster) Optimization of PVD SiCN Deposition for Cu/SiCN Hybrid Bonding Applications, ICEP-IAAC 2025, J. Choi, S. Jang, D. Lee, **H. Lee**, S. E. Kim

Tool Ownership and Training

S-fab (SeoulTech's cleanroom) Equipment Ownership *Mar. 2025 – Jan. 2027*

- Maintenance and Self-use training
- ICP CVD (Oxford, PlasmaPro 100), Aligner (EVG 610), Wafer Bonder (EVG 501), Shear Tester (MFM 1200 DGFT)

S-fab Process Training Assistant *Jan. – Feb. 2026*

- System Maintenance
- Equipment Troubleshooting
- SOP Standardization
- Facility Installation

Patents

Low-Temperature Hybrid Bonding Method and Hybrid Bonding Assembly Accordingly (US Patent Application Number, 19/365,396) Oct 2025

Skills Summary

Software/Tools: Origin, XPSPEAK4.1, JMP, Python, C, C++, Framer

Fabrication (Self-User): Sputter (SRN-100), ICP CVD (PlasmaPro 100), Aligner (EVG 610), Vacuum Oven, RIE etcher (PlasmaPro 80 Cobra65), Ion Milling (IM4000plus CTC), Mechanical Dicing (DISCO DAD3350), Stealth Dicing (DISCO DFL7341 SD), Die bonder (SET), Wafer Bonder (EVG 501), Probe Station, 3D profiler (Dektak 150), Nanospec-3000, Hall effect measurement (HMS-3000), MFM 1200 DGFT (Shear Test), Hitachi fineSAT (SAM).

Languages: Korean (Native), English (iBT TOEFL : 5.5 (110))

Reference

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